Hierarchical classification of breast cancer molecular subtypes from discrete aCGH copy-number profiles

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Abstract

Motivation: Breast cancer molecular subtypes (HER2+, HR+, Triple Negative) require distinct therapeutic strategies, yet classification from discrete array CGH copy-number profiles ($n = 100$, 2,834 regions) poses a high-dimensional, low-sample-size challenge. Wessels et al. (2005) showed that simple classifiers with univariate filtering match complex pipelines on continuous expression data, but whether this holds for discrete aCGH data with multi-class structure has not been tested.

Results: A 2×2 factorial experiment crossing two feature selectors (Kruskal-Wallis, Elastic Net) with two classifiers (Nearest Mean, Random Forest) revealed that Random Forest outperformed Nearest Mean in flat 3-class classification (BA = 0.791 vs 0.631). However, HER2+ recall was consistently high across all flat pipelines (0.764-0.998), suggesting that HR+ vs Triple Negative confusion was the true performance bottleneck. A hierarchical decomposition (Stage 1: HER2+ vs rest; Stage 2: HR+ vs TN) reversed the ranking, with Nearest Mean outperforming Random Forest in Stage 2. The best pipeline, hierarchical EN+NMC with plateau ensembling, achieved a combined 3-class balanced accuracy of 0.840 and an expected validation accuracy of 48 of 57 samples.

Availability: Code and predictions are provided in the submission archive.

1 Introduction

Breast cancer comprises molecular subtypes, including HER2-positive (HER2+), hormone receptor-positive (HR+), and triple-negative (TN), that differ in prognosis and require distinct therapeutic strategies (Perou et al., 2000; Sørlie et al., 2001). Array Comparative Genomic Hybridization (aCGH) enables genome-wide detection of DNA copy-number (CN) alterations; after segmentation and calling (van de Wiel et al., 2007, 2011), it produces high-dimensional discrete profiles suitable for classification. The present study classifies breast cancer subtypes from such CN profiles using 100 labelled training samples and 57 unlabelled validation samples, a typical high-dimensional, low-sample-size challenge.

Wessels et al. (2005) demonstrated that simple classifiers with univariate feature filtering match or outperform complex pipelines on continuous expression data (van 't Veer et al., 2002), and Haury et al. (2011) confirmed that univariate selection achieves competitive accuracy while offering better feature stability. However, these findings pertain to continuous data; whether they extend to discrete aCGH profiles and multi-class settings has not been tested. We therefore ask: does the Wessels principle hold for discrete CN data, and how do feature redundancy and multi-class structure affect classifier selection? To answer this, we employ a 2×2 factorial design comparing simple and complex feature selection and classification methods via nested cross-validation (CV), and explore a hierarchical decomposition motivated by the observation that HER2+ samples are trivially separable while HR+ vs TN confusion is the true bottleneck.

2 Methods

2.1 Data description

The training set comprises $n = 100$ samples profiled on a 244K-probe aCGH platform, summarized into 2,834 genomic regions with discrete CN calls: loss (-1), normal (0), gain (+1), or amplification (+2). Classes are HER2+ ($n = 32$), HR+ ($n = 36$), and TN ($n = 32$). A validation set of 57 unlabelled samples is provided. Coordinates use hg18/NCBI Build 36. The analysis was implemented in Python 3.10 with `scikit-learn` 1.7.2 (Pedregosa et al., 2011).

2.2 Label-free region merging

Adjacent genomic regions showed strong spatial autocorrelation (median Pearson $r = 0.92$ within 0.1 Mb).

Following the label-free CGHregions approach of van de Wiel and van Wieringen (2007), we merged adjacent regions with $r > 0.8$ by taking the median CN per sample, reducing the feature space from 2,834 to 273 segments (90.4% reduction). The threshold was chosen because 91% of adjacent pairs exceeded $r = 0.8$, and nested CV tests confirmed that omitting merging degraded performance by 7 percentage points (pp) ($p = 0.008$, Wilcoxon). Merging is entirely label-free and was performed before CV splits; the merging map was saved and applied to validation data at prediction time to ensure consistent segment boundaries. This is distinct from the label-dependent leakage described by Ambroise and McLachlan (2002).

2.3 Experimental design

Following Wessels et al. (2005), we adopted a 2×2 factorial design crossing two feature selectors with two classifiers (Figure 1).

Feature selection. The Kruskal-Wallis (KW) test (Kruskal and Wallis, 1952), a univariate method in the Wessels tradition, ranks features by their H-statistic across the three subtypes; the top $k \in \{5, 10, 15, 20, 30, 50, 75, 100\}$ are retained. As a multivariate alternative, Elastic Net (EN) multinomial logistic regression (Zou and Hastie, 2005) evaluates features jointly; the top $k \in \{5, 10, 20, 50\}$ (by summed absolute coefficients) are retained, with $C \in \{10^{-3}, \ldots, 10^{2}\}$ and `l1_ratio` $\in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ tuned in the inner CV loop.

Classifiers. The Nearest Mean Classifier (NMC), a simple linear method recommended by Wessels et al. (2005), assigns samples to the closest class centroid with optional shrinkage $\delta \in \{\text{None}, 0.1, 0.2, 0.5\}$. Random Forest (RF), a complex ensemble, uses 500 trees with balanced class weights (Breiman, 2001), tuning `max_features` $\in \{\text{sqrt}, 0.3\}$ and `min_samples_leaf` $\in \{1, 5\}$. Crossing the two selectors with two classifiers yields four pipelines with 32 (KW) or 800 (EN) inner CV configurations per fold.

2.4 Hierarchical classifier

HER2+ recall was near-perfect across RF pipelines (0.994-0.998; Table 1), while HR+ vs TN confusion was the bottleneck across all flat pipelines. A binary KW ranking on HR+ and TN samples surfaced chr12q and chr5q features (Figure 2B) instead of the chr17 ERBB2 signal dominating 3-class ranking, motivating a two-stage architecture.

Stage 1 classifies HER2+ vs rest using a fixed KW+RF model with $k = 5$ chr17q12 features at the ERBB2 amplicon ($H = 73.9$, $p = 9 \times 10^{-17}$), achieving BA = 1.000 in all 1,000 outer folds. Stage 2 classifies the remaining samples as HR+ or TN using one of the four pipeline variants with full inner CV tuning on non-HER2+ samples only, accessing the chr12q/chr5q signal that 3-class ranking obscures (Figure 1).

2.5 Cross-validation scheme

We employed nested, stratified, repeated 5-fold CV (Wessels et al., 2005; Ambroise and McLachlan, 2002). The outer loop estimates generalisation performance (50 repeats $\times$ 5 folds = 250 evaluations for flat; 200 $\times$ 5 = 1,000 for hierarchical). The inner 5-fold loop tunes hyperparameters, with all feature selection performed strictly within inner folds to prevent leakage. Performance is measured by balanced accuracy (BA), the arithmetic mean of per-class recalls (Brodersen et al., 2010). For hierarchical models, we report both the combined 3-class BA and Stage 2 BA (BA2), the latter computed only on samples routed to Stage 2 (HR+ vs TN). The workflow is shown in Figure 1.

2.6 Plateau ensembling

Inner CV folds in Stage 2 contain only ~11 samples, insufficient to reliably rank 800 EN configurations: 303 distinct top-1 selections appeared across 1,000 folds, with negative inner/outer-test correlation ($r = -0.378$). Plateau ensembling addresses this by retaining all configurations within one pooled SD of the best mean inner-CV score (up to 15 members) and averaging their predicted class probabilities. The one-SD threshold was fixed a priori. Leave-one-repeat-out analysis confirmed plateau robustness: on average, fewer than 1 of 15 members changed when any single repeat was omitted, indicating that plateau membership is stable under resampling rather than overfit to specific fold compositions.

2.7 Statistical comparison and final model

Pipeline differences were assessed using the Nadeau-Bengio corrected t-test (Nadeau and Bengio, 2003), which accounts for non-independence in repeated CV, and bootstrap confidence intervals (CIs; 10,000 resamples) (Bouckaert and Frank, 2004). The final submitted model (EN+NMC with plateau ensembling) was retrained on all 100 samples with fixed plateau membership and the same region-merging map applied to validation data.

3 Results

3.1 Flat 2x2 experiment

In the flat setting, RF outperformed NMC regardless of feature selector: KW+RF achieved BA = 0.791 vs 0.631 for KW+NMC (Table 1). HER2+ recall was high across all pipelines (0.764-0.998), but TN recall ranged from only 0.507 to 0.641 (Figure 3B, left), identifying HR+ vs TN confusion as the bottleneck.

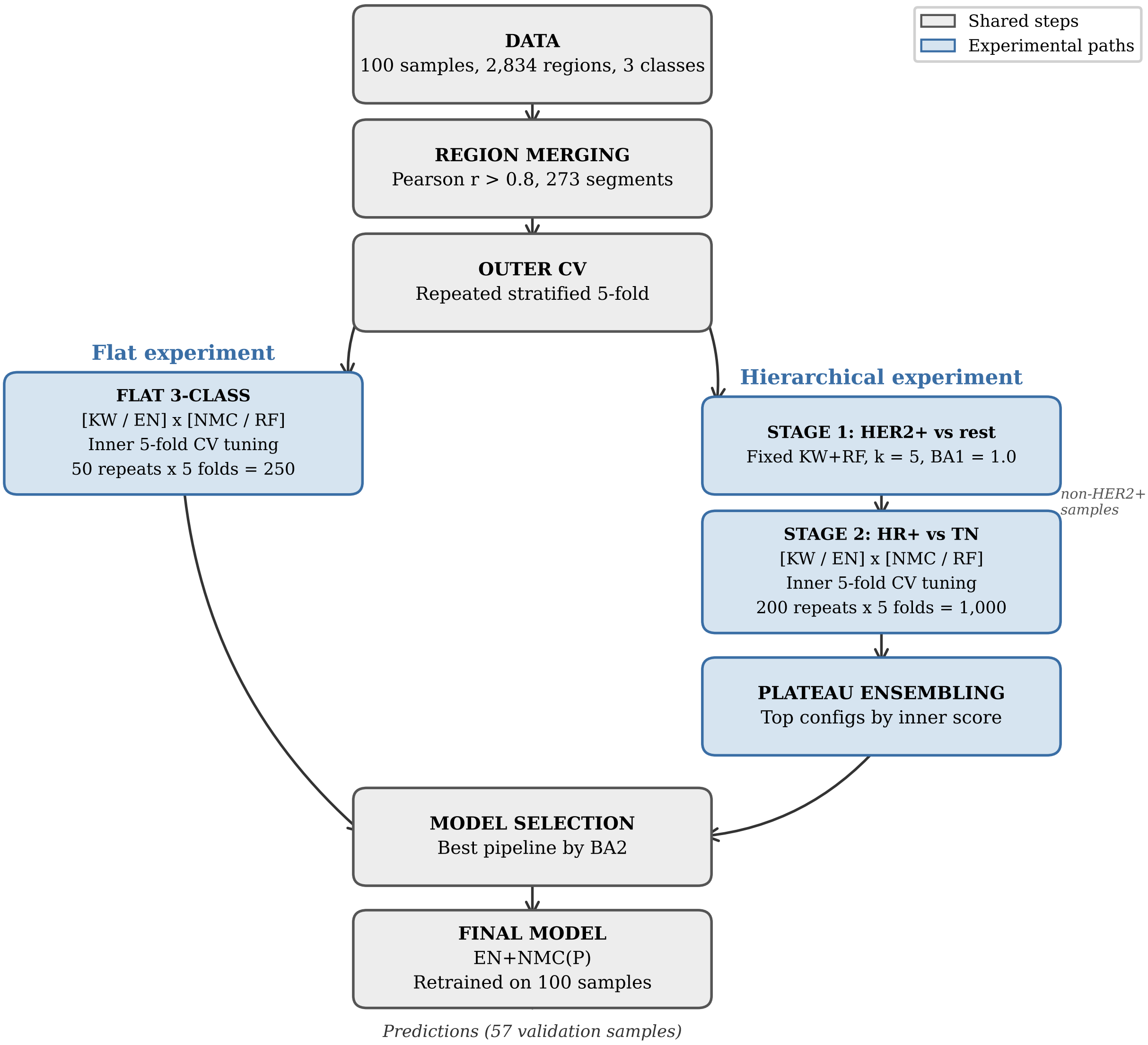

Figure 1: Nested CV workflow. The outer loop (200 repeats $\times$ 5 folds) estimates performance; the inner loop tunes hyperparameters. Stage 1 separates HER2+ (BA1 = 1.000); Stage 2 classifies HR+ vs TN. Feature selection occurs inside each inner fold.

Table 1: Performance comparison across the 2×2 experimental design. Flat pipelines classify all three subtypes simultaneously, while hierarchical pipelines first separate HER2+ from others (BA1 = 1.0 for all) and then distinguish HR+ from TN. Plateau ensembling (P) averages predictions from top-performing hyperparameter configurations. Per-class columns report recall; all metrics are presented as mean $\pm$ SD across outer folds. Med. k indicates median number of selected features; bold values highlight best performance per column.

Experiment	Pipeline	BA	BA2	HER2+	HR+	TN	Med. k
Flat	KW+NMC	0.631±0.031	–	0.764±0.069	0.623±0.064	0.507±0.079	15
	KW+RF	0.791±0.025	–	0.994±0.014	0.736±0.056	0.641±0.056	75
	EN+NMC	0.746±0.032	–	0.909±0.035	0.698±0.047	0.630±0.066	10
	EN+RF	0.770±0.029	–	0.998±0.008	0.696±0.064	0.616±0.058	20
Hier. (base)	KW+NMC	0.824±0.031	0.736±0.046	**1.000**±0.000	0.760±0.056	0.712±0.068	30
	KW+RF	0.814±0.029	0.721±0.044	**1.000**±0.000	0.748±0.058	0.692±0.057	30
	EN+NMC	0.819±0.026	0.729±0.040	**1.000**±0.000	0.729±0.051	0.730±0.058	20
	EN+RF	0.794±0.031	0.691±0.046	**1.000**±0.000	0.726±0.058	0.655±0.070	50
Hier. (plateau)	KW+NMC(P)	0.834±0.026	0.750±0.040	**1.000**±0.000	0.773±0.047	0.727±0.064	100
	EN+NMC(P)	**0.840**±0.022	**0.759**±0.032	**1.000**±0.000	0.777±0.045	**0.741**±0.047	81
	Ensemble	0.838±0.025	0.757±0.037	**1.000**±0.000	**0.778**±0.046	0.735±0.056	–

3.2 Hierarchical classifier

The hierarchical decomposition reversed the classifier ranking: NMC outperformed RF in Stage 2 (HR+ vs TN), with bootstrap CIs excluding zero for all NMC-vs-RF pairs. EN+NMC(P) achieved the best BA = 0.840 and BA2 = 0.759 (Table 1; Figure 3). The Nadeau-Bengio test found no significant pairwise differences, consistent with its conservative nature for repeated CV.

3.3 Plateau ensembling

Plateau ensembling reversed the classifier ranking: base EN+NMC (BA2 = 0.729) trailed both KW+NMC (0.736) and standalone EN (0.738) because EN's 800-configuration grid overfits on $\sim$11 inner-fold samples (negative inner/outer correlation; Section 2.6), whereas KW's 32 configurations remain stable. By averaging near-optimal configurations rather than selecting the single overfit best, plateau ensembling neutralized this: EN+NMC(P) (0.759) surpassed KW+NMC(P) (0.750) and standalone EN(P) (0.741; Table 1). Yet overfitting alone does not explain the result: standalone EN(P) accesses the same 800 configurations but gains only +0.3 pp, while EN+NMC(P) gains +3.0 pp. The critical difference is the classifier. Each plateau member selects a different feature subset, producing a different NMC with centroids in a distinct subspace. Averaging their predictions is analogous to random subspace ensembling: the consensus across multiple complementary views of the data is more robust than any single projection, effectively transforming NMC from a single centroid classifier into a multi-view ensemble. EN's logistic regression, by contrast, uses regularization to implicitly weight all features, so different hyperparameter configurations converge to similar decision boundaries and averaging adds little diversity.

3.4 Single best biomarker region

The chr17q12 ERBB2 amplicon (KW $H = 73.9$, $p = 9 \times 10^{-17}$) enabled perfect Stage 1 separation using five features, consistent with its established role as a clinical biomarker (Slamon et al., 1987). For Stage 2, discriminative signal resided on chr12q and chr5q (Figure 2).

3.5 Accuracy estimate

The expected number of correct predictions was estimated as $\sum_c \hat{r}_c \cdot (n_c/N) \cdot 57$, where $\hat{r}_c$ is the mean per-class recall across 200 repeats and n_c/N the training class proportion. With HER2+ = 1.000, HR+ = 0.777, TN = 0.741, this yielded 47.7 $\approx$ 48. Sensitivity analysis under alternative class proportions produced estimates of 46-48; 48 was submitted.

3.6 Iterative model refinement

Three methodological refinements improved accuracy beyond the initial draft submission. (1) The flat KW+RF (BA = 0.791) was replaced by a hierarchical decomposition, improving the best BA to 0.824 (+3.3 pp) and reversing the classifier ranking: NMC outperformed RF in Stage 2, consistent with the Wessels principle on a well-structured subproblem. (2) Plateau ensembling raised EN+NMC from BA = 0.819 to 0.840 (+2.1 pp) and BA2 from 0.729 to 0.759 (+3.0 pp). (3) The estimate was updated from BA $\times$ 57 to a per-class weighted formula, increasing from 45 to 48.

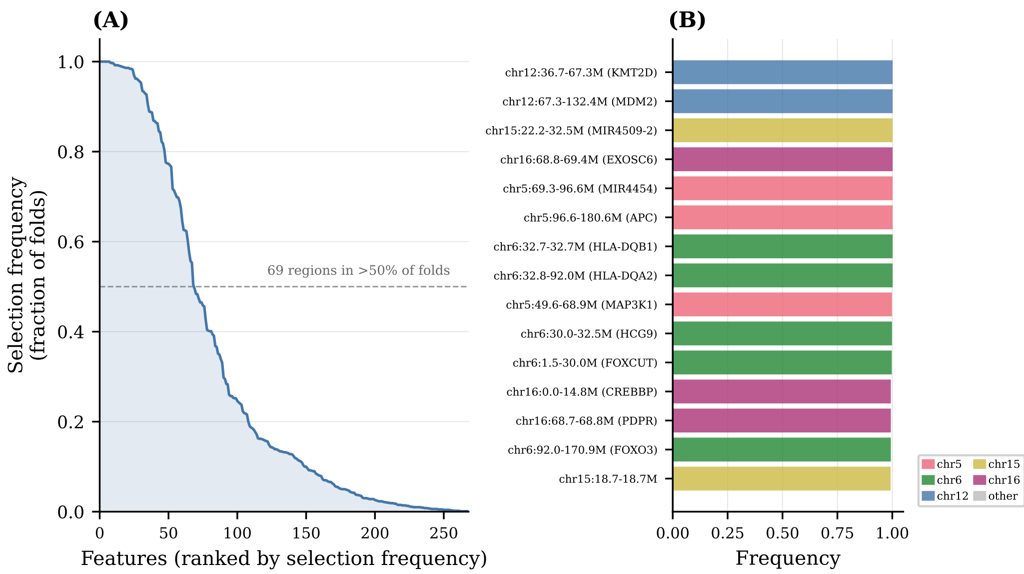

Figure 2: Feature selection stability for EN+NMC(P) Stage 2. **(A)** Selection frequency across 1,000 outer folds for all 273 segments. **(B)** Top 15 regions, color-coded by chromosome. Chr12 and chr5 dominate, distinct from the chr17 ERBB2 signal in Stage 1.

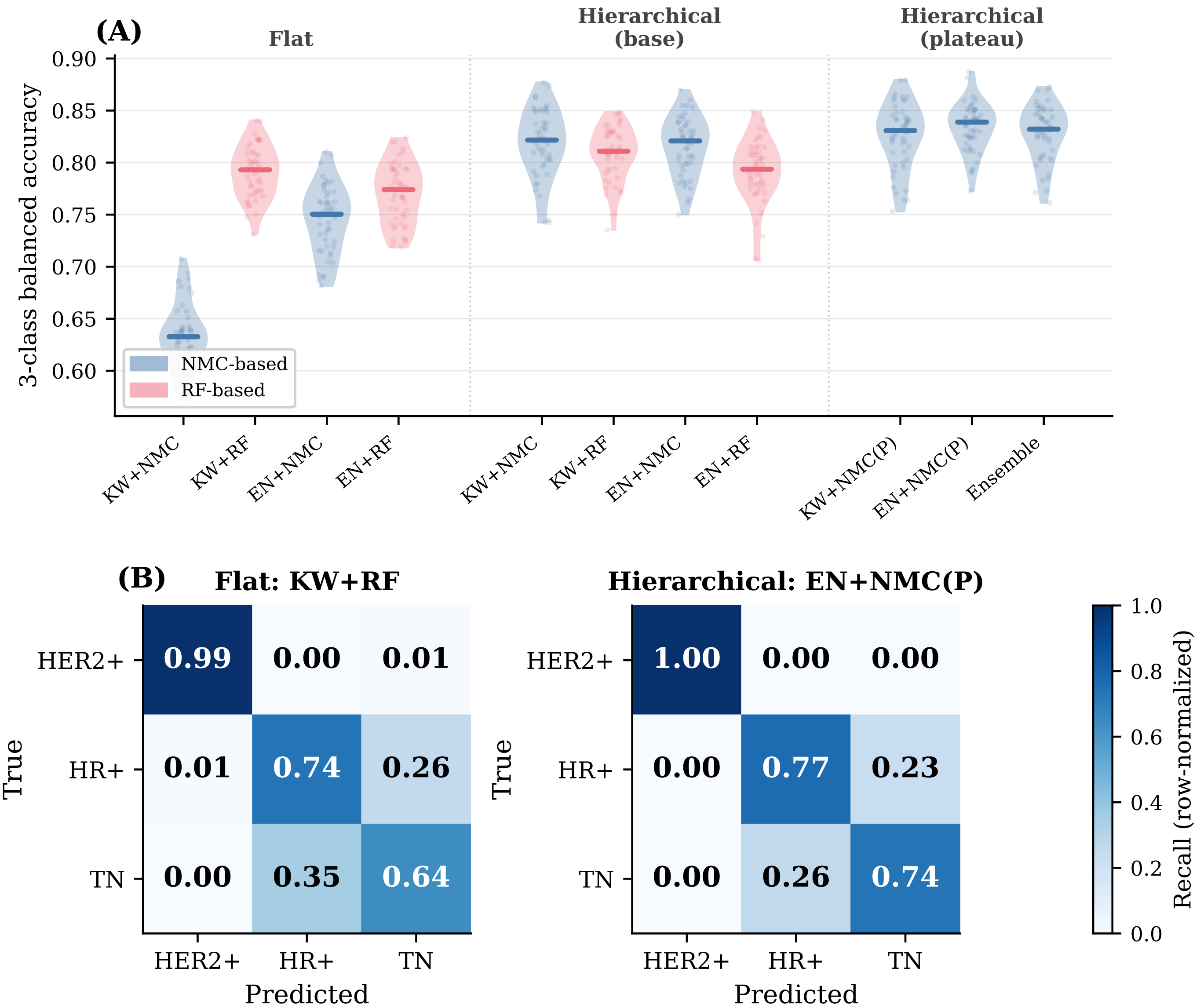

Figure 3: Model comparison. **(A)** BA distributions per pipeline (50 repeats flat, 200 hierarchical). All hierarchical variants outperform the best flat pipeline. **(B)** Confusion matrices for KW+RF (flat, left) and EN+NMC(P) (hierarchical, right), pooled over all outer folds.

4 Discussion

The Wessels principle is conditional on problem structure. In the flat 3-class setting, RF outperformed NMC (BA = 0.791 vs 0.631; Figure 3A), apparently contradicting findings on continuous expression data. After hierarchical decomposition removed the trivially separable HER2+ class, NMC outperformed RF in the binary HR+ vs TN problem. The HER2+ class acted as a confounder: its perfect separability inflated RF's advantage in the 3-class setting. In both settings, feature selector choice mattered less than classifier choice, consistent with Wessels et al. (2005). The discovered biomarker regions harbor genes with established roles in breast cancer biology: chr17q12 contains *ERBB2* (HER2 receptor tyrosine kinase), while Stage 2 signals map to *CDK4* and *MDM2* on chr12q (cell cycle regulation and p53 antagonism) and *PIK3R1* on chr5q (PI3K/AKT pathway activation), consistent with known molecular differences between luminal (HR+) and basal-like (TN) subtypes (Chin et al., 2006; Sørlie et al., 2001).

Limitations include the small sample size ($n = 100$, 68 for Stage 2) and discrete CN representation. Approximately 10 samples were consistently misclassified (>80% across 1,000 test appearances; Figure 3B), suggesting the BA2 $\approx$ 0.759 ceiling reflects irreducible noise rather than classifier capacity. Future work could integrate expression data for multi-omic classification and test whether the hierarchical NMC advantage generalises to larger cohorts. In conclusion, simple classifiers can outperform complex ones on discrete aCGH data, but only when multi-class structure is decomposed to match the underlying biology.

Author contributions

C. Botos: EDA, region merging, hierarchical nested CV, plateau ensembling, cpu-cluster. **A. Wagner:** Flat nested CV, KW+NMC, KW+RF, research question conceptualization. **A. Michailidis:** Flat nested CV, EN+NMC, EN+RF, standalone EN. **Y. Qiao:** Final training, estimate computation. All authors contributed to intellectual design and report writing.

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